

Alvin Isac

Robotics Engineer | UAV Control | Autonomous Systems

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Nantes, France | Work Authorization: APS France | Available Immediately

Robotics Engineer specialized in UAV control, autonomous systems, and ROS2-based robotic software. Experienced in fault-tolerant NMPC, multi-UAV payload transportation, ROS1-to-ROS2 migration, and autonomous navigation across real and simulated robotic platforms. Strong background in C++, Python, MATLAB, PX4, Gazebo, and optimization-based control for robotics and aerial systems.



Education

École Centrale de Nantes & University of Genoa

European Master's in Advanced Robotics | Erasmus Mundus

Aug 2023 – Oct 2025

Nantes, France – Genova, Italy

- Selected as one of the two students for the EMARO program
- Graduated from double-degree master's program across École Centrale de Nantes, France and the University of Genoa, Italy, two leading European institutions for robotics and automation.
- *Courses(M1 & M2)*: Robot Dynamics & Control, Classical Linear Control, Optimization Techniques, Signal Processing, Advanced Robot Programming, Computer Vision, Machine Learning, Artificial Intelligence
- **Grade: 8.8 / 10 CGPA**

Anna University

B.E. Mechanical Engineering

Aug 2017 – May 2021

Chennai, India

- *Courses*: Control Engineering, Kinematics of Machinery, CAD/CAM, Fluid Mechanics
- **Grade: 7.5 / 10 CGPA**

Experience

Genoa Robotics and Automation Laboratory (GRAAL)

Graduate Researcher | Master's Thesis Research | [GitHub](#)

Feb 2025 – Oct 2025

Genoa, Italy

- **Thesis**: [Aggressive Maneuver Planning and Control of a Multi-UAV and Payload System after a Failure](#)
- **Supervisors**: Prof. Marco Baglietto, Prof. Olivier Kermorgant, and Dr. Andrea Delbene
- *Tools*: ROS2, PX4, QGroundControl(QGC), C++, Python, MATLAB/Simulink, Gazebo, CasADi, IPOPT
- Designed a two-phase Nonlinear Model Predictive Control framework for cooperative UAV payload transport under mid-flight failure scenarios, separating aggressive post-failure stabilization from precision trajectory tracking.
- Developed a nonlinear UAV-payload-cable dynamic model including quadrotor dynamics, payload motion, and spring-damper cable tension behavior for oscillation suppression and control design.
- Formulated NMPC cost functions, safety constraints, and phase-switching logic to suppress payload oscillations, preserve formation coordination, and maintain stable operation after failure.
- Implemented and tuned the controller in MATLAB/Simulink and integrated the robotics stack in ROS2/PX4/Gazebo, using CasADi and IPOPT for nonlinear optimization.
- Evaluated the controller under fault-injection scenarios, demonstrating rapid stabilization (~ 3 s), smooth transition to tracking, and robust formation maintenance.

Technical Projects

ROS1-to-ROS2 Migration of TurtleBot2 Mobile Robot [GitHub](#)

May 2024

- *Tools*: ROS1, ROS2, C++, Python, URDF/Xacro, RViz2, Kobuki, Hokuyo, Kinect
- Migrated a TurtleBot2 Kobuki platform from ROS1 to ROS2 Humble, including driver integration, launch workflows, and ROS2 interface adaptation.
- Built a real-robot bring-up pipeline with Hokuyo LiDAR and Microsoft Kinect integration for ROS2-based teleoperation, odometry, IMU, and sensor streaming.
- Developed launch files that configured the Kobuki ROS2 node, loaded runtime parameters, and remapped command topics to standardized interfaces.
- Used `robot_state_publisher`, `joint_state_publisher`, and RViz2 to validate the robot model and sensor-frame alignment on the physical platform.

ROS2 Autonomous Navigation and ArUco Marker-Based Decision Making [GitHub](#)

Jan 2025

- Tools: ROS2 Humble, Nav2, Gazebo, Python, OpenCV, Xacro, Slam Toolbox
- Built an autonomous mobile robot workflow in ROS2 and Gazebo, integrating Nav2 navigation with ArUco marker perception for task-driven decision making.
- Developed a Python control node using `nav2.simple_commander` and BasicNavigator to send the robot to multiple waypoints and manage mission execution in the map frame obtained through Slam Toolbox.
- Processed ArUco marker detections from the relevant topic to extract marker IDs and link perception outputs with corresponding navigation waypoints.
- Implemented decision logic that compared observed markers across multiple goals and autonomously commanded the robot to navigate to the location with the lowest marker ID.

Biomedical Robotics: EMG Processing and PHANTOM Haptic Robot Control [GitHub](#)

Sep 2024

- Tools: MATLAB, Simulink, VRML, PHANTOM Robot, EMG Signal Processing
- Developed biomedical robotics projects combining EMG signal processing and real-time control of a 3-DOF PHANTOM haptic device for rehabilitation-oriented applications.
- Built an EMG preprocessing pipeline using 30–450 Hz bandpass filtering, rectification, smoothing, normalization, and threshold-based peak extraction for multi-channel muscle signals.
- Implemented Jacobian and Jacobian-transpose formulations for device kinematics and control in MATLAB.
- Integrated the PHANTOM device with Simulink through custom functions to read joint angles, Cartesian position, velocity, and transformation matrices while sending torque commands in real time.

Multi-Class Animal Image Classification with Classical Machine Learning [GitHub](#)

Dec 2024

- Tools: Python, Google Colab, scikit-learn, scikit-image, NumPy, Matplotlib
- Built an end-to-end image classification pipeline for 10 animal classes using the Animals dataset, covering data loading, preprocessing, training, and evaluation.
- Preprocessed raw images through grayscale conversion, 100x100 resizing, and flattened feature extraction for classical machine learning models.
- Implemented and compared SVM and KNN classifiers in scikit-learn, applying GridSearchCV and structured experiments across kernels and model settings.
- Evaluated models using classification reports, ROC-based analysis, confusion matrix review, and cross-validation to assess class-level predictive performance.

Skills

Programming Languages: C++, Python, MATLAB, Bash, C

Robotics & Control: ROS2, PX4, Gazebo, Nav2, Fault-Tolerant Control, Formation Control, Trajectory Tracking, PID, LQR, NMPC

Robot Modeling & Perception: URDF, Slam Toolbox, Xacro, TF, RViz2, LiDAR Integration, OpenCV

Optimization & ML: CasADi, IPOPT, scikit-learn, NumPy, Matplotlib, SVM, KNN

Development & Cloud Tools: Git, Docker, n8n automation, VS Code, Google Cloud Console, OpenRouter API, Tailscale, VNC

Soft Skills: Independent & Self-Driven, Problem Solving, Fault Diagnosis & Recovery, Attention to Detail

Training & Certifications

Control System UAV (3D Dynamics & Control)	Udemy – Sep 2025
Autonomous Cars: Math + PID + MPC	Udemy – Sep 2025
ROS2 SLAM & Navigation	Udemy – Nov 2024
ROS2 Foxy & Humble	Udemy – Apr 2024
ROS2 TF, URDF, RViz & Gazebo	Udemy – Jan 2024
Control Design with Simulink	Matlab – Nov 2024
Machine Learning Specialization	Coursera – Jan 2024
Supervised Machine Learning: Regression and Classification	Coursera – Dec 2023
Complete CATIA V5 Training	Udemy – Feb 2023

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Languages

English (Fluent), French (Intermediate), Tamil (Native)

Honours & Awards

Consortium Scholarship for EMARO Student 5,000 EUR recipient	2024
1st Place in University Level Chess Tournament	2021